DEVOPS SAFEGUARD – WEEKLY PROJECT LOG

Final-Year Project | Six Delivery Phases: Weekly Tasks Planned and Achieved

Week	Task Planned	Task Achieved
Week 1 FOUNDATION	Finalise the project direction, define the system architecture, and set up the Next.js, Supabase, and PostgreSQL foundation.	Built the marketing site, email/password authentication with OTP verification, and an RLS-scoped project workspace. Applied the core schema, tenant model, and row-level security policies.
Week 2 INCIDENTS	Design and implement the end-to-end incident lifecycle: observe, detect, investigate, evidence, analyse, approve, remediate, and verify.	Shipped incident, service, timeline, evidence, AI-analysis, and remediation tables with RLS. Built the incident detail view, print report, and JSON export. AI remains read-only and every remediation is human-approved.
Week 3 ONBOARDING	Build a guided project-onboarding wizard with automated repository analysis for Docker, Compose, Kubernetes, Helm, and application frameworks.	Delivered a **10-step wizard** with GitHub analysis, evidence paths, confidence tiers, and canonical configuration. Runtime, dependency, and health settings now read back correctly on the dashboard.
Week 4 MONITORING	Replace manual health checks with an unattended scheduler that applies failure thresholds and requires verified recovery.	Built a standalone Node worker with Postgres-backed claims, leases, retries, and graceful restarts. Added threshold-gated incident creation, **3-consecutive-healthy** recovery, and honest next-check and staleness states.
Week 5 VALIDATION	Validate the complete pipeline against a real deployed application and harden the security and concurrency boundaries.	Drove Podinfo on AWS/K3s through a sustained HTTP 500: one incident at threshold, evidence-backed diagnosis, verified recovery, and resolution. Added default-deny private-target support and database deduplication guards.
Week 6 DEPLOYMENT	Deploy the platform to production, upgrade dependencies for security, refine the monitoring and authentication UX, and complete release QA.	Shipped the Vercel production build on Next.js 16.4.2 with editable health monitoring, a redesigned OTP and day/night auth experience, and responsive polish. **242** unit tests, typecheck, lint, and production build pass.

Note: Each week corresponds to one delivery phase. The Task Achieved column reflects work completed and verified in the repository.